

# SPRFusion: Semantic and Physics-driven Robust Radar-Camera Fusion for 3D Object Detection

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## Abstract

Radar-Camera fusion has emerged as a promising paradigm for 3D object detection in autonomous driving. While millimeter-wave radar provides valuable Doppler velocity and reliable radial depth, its point clouds are inherently sparse and noisy. Existing methods often naively adapt LiDAR-Camera fusion architectures, overlooking the profound semantic-geometric asymmetry between modalities and failing to fully exploit radar's physical kinematics. To address these limitations, we propose SPRFusion, a semantic and physics-driven robust Radar-Camera fusion framework. Rather than relying solely on visual cues, SPRFusion explicitly leverages radar Doppler priors to guide the cross-frame dynamic alignment of camera BEV features, effectively mitigating spatiotemporal misalignment for highly dynamic objects. To handle the distance-dependent detection difficulty caused by radar sparsity, SPRFusion formulates a difficulty-aware circular query distribution strategy, dynamically concentrating limited computational resources on highly challenging zones. Moreover, instead of treating radar merely as a spatial anchor, SPRFusion injects open-world physical attribute priors derived from Large Language Models (LLMs) into the sparse geometric signals, achieving explicit semantic calibration. Through an uncertainty-aware gating mechanism, SPRFusion dynamically refines cross-modal fusion weights, significantly enhancing anti-noise robustness under severe environmental degradation. Extensive experiments on the nuScenes and View-of-Delft benchmarks show strong performance. With a ResNet-50 backbone, SPRFusion achieves 62.4% NDS and 55.4% mAP on the nuScenes validation set, and yields a highly competitive 68.2% NDS on the test set without employing future frames, test-time data augmentation, or model ensembling, demonstrating its effectiveness as a robust multimodal 3D detector.

## CCS Concepts

• **Computing methodologies** → **3D object detection**; *Multimodal systems*; *Sensor fusion*.

## Keywords

autonomous driving, 3D object detection, robust multimodal fusion

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**Figure 1: Representative adverse scenarios show that existing radar-camera fusion methods often produce unstable responses and missed detections, while SPRFusion yields more concentrated and reliable activations around object regions, highlighting the need for robust cross-modal fusion under sensor degradation.**

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## 1 Introduction

Precise 3D object detection is a fundamental requirement for autonomous driving. Compared with costly LiDAR-based systems, radar-camera fusion offers a more cost-effective perception solution while retaining complementary sensing properties: cameras provide rich semantic cues, whereas millimeter-wave radar offers reliable radial depth and Doppler velocity [13, 30]. Recent studies have demonstrated the potential of radar-camera fusion for 3D detection [12, 21, 34, 40]. However, due to the substantial heterogeneity between the two modalities, existing methods still face notable limitations in robust spatio-temporal modeling. [6, 12, 20, 21, 23]

Most existing methods perform fusion either in a unified Bird's-Eye-View (BEV) space or through query-based cross-view interaction [6, 20, 23]. Although effective, these paradigms still exhibit notable limitations in robust spatio-temporal perception. In practice, temporal aggregation predominantly relies on ego-motion compensation or implicit history modeling, which is insufficient to handle fast-moving objects with independent motion, leading to severe cross-frame misalignment. Meanwhile, the commonly adopted uniform query initialization fails to reflect the highly non-uniform distribution of detection difficulty across driving scenes, resulting

in suboptimal allocation of computational resources. More critically, existing fusion strategies typically perform naive feature interaction without explicitly addressing the intrinsic semantic gap between sparse radar signals and dense visual representations, nor the reliability variation introduced by sensor degradation. As illustrated in Figure 1, under challenging open-world conditions—such as rain, nighttime, and heavy occlusion—conventional fusion paradigms are often compromised by degraded modalities. This leads to scattered predictions and frequent missed detections, highlighting the urgent need for a more robust and reliability-aware fusion paradigm.

To address these challenges, we propose SPRFusion, a semantic- and physics-driven radar-camera fusion framework for robust 3D object detection. Instead of relying on heuristic fusion pipelines, SPRFusion is built upon three tightly coupled designs that explicitly tackle temporal misalignment, spatial sampling inefficiency, and cross-modal inconsistency. At the temporal level, we introduce a Cross-Modal Temporal BEV Alignment module (CMTA), which leverages radar Doppler cues as explicit motion priors to guide the alignment of historical camera BEV features, thereby improving temporal consistency for highly dynamic objects. From a spatial perspective, we further develop a Difficulty-Aware Circular Distribution strategy (DACD), which adaptively allocates object queries according to the non-uniform distribution of detection difficulty, enabling more effective utilization of computational resources in challenging regions. At the feature interaction stage, we design a dual-robust cross-modal decoder that performs radar-aware semantic calibration while dynamically adjusting fusion weights based on modality reliability, leading to more stable and noise-resistant predictions under adverse conditions.

We conducted systematic experimental evaluations on two authoritative benchmarks, namely nuScenes and View-of-Delft (VoD). Utilizing ResNet-50 as the backbone network, SPRFusion achieves state-of-the-art (SOTA) performance on the nuScenes validation set with 62.4% NDS and 55.4% mAP. Furthermore, it reaches a highly competitive result of 68.2% NDS on the test set, effectively validating the efficacy of our proposed method in multi-modal 3D detection. In summary, the primary contributions of this paper are as follows:

- We propose a Doppler-conditioned cross-modal temporal alignment module (CMTA), which leverages radar Doppler cues to guide the warping of historical camera BEV features, improving temporal consistency for dynamic objects.
- We design a Difficulty-Aware Circular Distribution (DACD) strategy that models object occlusion, image degradation, and radar sparsity as a continuous spatial difficulty field in polar coordinates. By adjusting the radial density of queries, it enables more efficient feature sampling in high-difficulty regions.
- We construct a dual-robust cross-modal decoder that incorporates LLM-derived semantic priors as a static codebook to enhance radar feature representation, and integrates an uncertainty-aware gated fusion mechanism to dynamically suppress noise and reconcile cross-modal discrepancies.
- Extensive experiments show that SPRFusion achieves competitive detection accuracy while improving robustness under challenging conditions such as rain and nighttime, demonstrating its effectiveness in real-world multimodal perception.

## 2 Related Work

### 2.1 Radar-Camera 3D Object Detection

Recent radar-camera fusion approaches primarily rely on BEV or Query-based architectures. Within the BEV paradigm, methods like CRN [13] and RCBEVDet [21] leverage attention mechanisms for feature fusion. To mitigate the depth ambiguity during view transformation, HyDRa [40] and CRAB [15] explicitly inject radar depth or occupancy priors. Query-based methods utilize 3D queries for cross-view extraction. CRAFT [12] achieves proposal-level early fusion, while RaCFormer [6] explores polar initialization. Recent studies like RCTrans [18] and RaCFusion [39], further optimize query generation and sequence decoding. To address modality asymmetry, research focuses on heterogeneous feature alignment [1, 24, 53], or geometric consistency for long-range targets [5]. Furthermore, to handle cross-frame misalignment of dynamic objects, models introduce dynamic depth compensation such as R4Det [43], object-level kinematic modeling Stream-PETR [38], or temporal knowledge distillation RCTDistill [2]. Diverging from existing appearance-based or implicit modeling, this paper reconstructs fusion from dual spatio-temporal dimensions: spatially, via a Difficulty-Aware Circular Distribution (DACD) for optimized sampling; temporally, by leveraging radar Doppler to explicitly guide displacement field learning (CMTA) for kinematic compensation.

### 2.2 Uncertainty Modeling and Robust Perception

Sensor degradation triggers severe performance drops in multi-modal fusion, as evidenced by benchmarks such as BEVRobust [44] and RoboFusion [36]. To enhance baseline robustness [20, 23], which often suffers from implicit memory reliance, recent studies explore feature augmentation via Cross-Modal Knowledge Distillation (KD), leveraging LiDAR priors [29], Vision Foundation Models [28], and hierarchical decoupling [33]. In parallel, explicit uncertainty estimation is adopted to quantify depth variance [27], predict BEV reliability distributions [8, 50], and perform feature denoising [48]. However, visual-centric enhancement methods relying on image quality remain limited under severe physical occlusion. To address this limitation, we design a decoupled uncertainty-aware cross-modal gating mechanism that maintains robust adaptability under severe modal degradation.

### 2.3 Foundation Models in Autonomous Driving

Large Language Models (LLMs) and Vision-Language Models (VLMs) are increasingly integrated into autonomous driving to enhance 3D reasoning. Existing research primarily focuses on high-level cognition and decision-making, including map-based scene description [3, 37], logical reasoning [47, 51, 52], and 3D geometric understanding [46]. However, few studies exploit LLM-derived priors to guide low-level multimodal feature aggregation. To this end, we propose using LLM-generated physical attributes as a static semantic codebook to calibrate sparse radar features, thereby effectively reconciling semantic discrepancies across modalities.

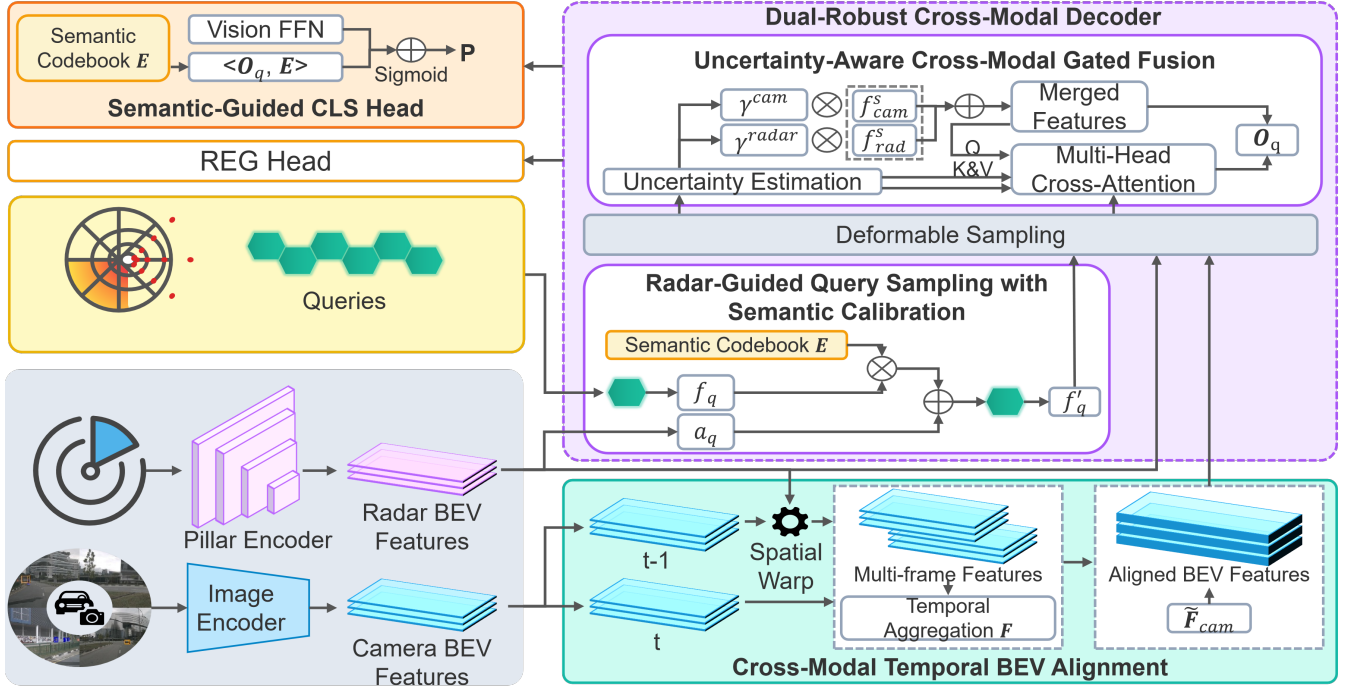


Figure 2: Overall architecture of the proposed SPRFusion framework. The framework consists of modality-specific BEV encoders, the proposed Cross-Modal Temporal BEV Alignment (CMTA) module, the Difficulty-Aware Circular Distribution (DACD) strategy for query initialization, and a dual-robust cross-modal decoder for semantic calibration and uncertainty-aware cross-modal fusion, followed by classification and regression heads for 3D detection.

### 3 Method

#### 3.1 Overall Framework

As illustrated in Figure 2, SPRFusion is a dual-driven radar-camera fusion framework for 3D object detection, guided by semantic and physical priors. The framework consists of modality-specific encoders, a Cross-modal Temporal BEV Alignment (CMTA) module, a Difficulty-Aware Circular Distribution (DACD) module, and a Dually Robust Cross-modal Decoder. In the feature extraction stage, the image encoder extracts multi-scale perspective features, which are projected into camera BEV features via view transformation, while the radar encoder processes point clouds into radar BEV features. The camera BEV features are then processed by CMTA, which leverages radar Doppler as a motion prior to predict a residual warping field for non-rigid alignment of historical features. Before cross-modal interaction, Object Queries serve as the medium for multimodal feature extraction. Through DACD, queries adaptively adjust their radial density in a polar coordinate system based on a spatial difficulty field, allocating more resources to challenging regions. Finally, the queries are fed into the Dually Robust Cross-modal Decoder, which performs semantic calibration using a static LLM-based codebook and applies uncertainty-aware gated fusion to suppress degraded modalities. The fused query representations are used for final classification and regression.

#### 3.2 Cross-Modal Temporal BEV Alignment (CMTA)

In BEV-based 3D object detection, temporal aggregation mitigates occlusion and enables motion estimation [11, 23]. Existing methods rely on ego-motion for rigid alignment, which fails for independently moving objects, causing spatial misalignment and ghosting artifacts, as shown in Figure 3(a). To address this, we propose CMTA, which uses radar Doppler as a motion prior to estimate a dense displacement field for aligning dynamic objects across frames, as shown in Figure 3(b).

Specifically, let  $\mathbf{F}_{cam}^{t-1} \in \mathbb{R}^{C \times H \times W}$  denote the historical camera BEV feature map at time step  $t-1$ , and  $\mathbf{F}_{rad}^t \in \mathbb{R}^{C' \times H \times W}$  denote the radar BEV feature map at time step  $t$ . The channel dimension of  $\mathbf{F}_{rad}^t$  encodes motion information after projection onto the BEV plane.

We concatenate the historical camera features and the current radar features along the channel dimension and feed them into a displacement field estimator  $\mathcal{H}$  parameterized by a lightweight residual convolutional network. This network predicts a two-dimensional motion compensation vector  $\Delta \mathbf{p} \in \mathbb{R}^{2 \times H \times W}$  for each spatial coordinate  $\mathbf{p} = (x, y)$ :

$$\Delta \mathbf{p} = \mathcal{H}(\text{Concat}(\mathbf{F}_{cam}^{t-1}, \mathbf{F}_{rad}^t)). \quad (1)$$

Physically, this dense displacement field represents the dynamic displacement vector of each spatial grid over the time interval  $\Delta t$ .

After obtaining the displacement field, we use a grid sampling mechanism to perform non-rigid spatial distortion on the historical camera features. To prevent over-compensation in static regions, we introduce a position-dependent modulation coefficient  $\alpha(\mathbf{p}) \in [0, 1]$  to scale the displacement, thereby obtaining the compensated sampling location:  $\mathbf{p}' = \mathbf{p} + \alpha(\mathbf{p}) \cdot \Delta\mathbf{p}(\mathbf{p})$ . Since  $\mathbf{p}'$  typically represents a continuous fractional coordinate, we apply bilinear interpolation to aggregate the features of its adjacent grids, yielding the spatially aligned historical feature  $\hat{\mathbf{F}}_{\text{cam}}^{t-1}$ . This process can be formulated as:

$$\hat{\mathbf{F}}_{\text{cam}}^{t-1}(\mathbf{p}) = \sum_{\mathbf{q} \in \mathcal{N}(\mathbf{p}')} \mathbf{F}_{\text{cam}}^t(\mathbf{q}) \max(0, 1 - |x' - x_q|) \max(0, 1 - |y' - y_q|), \quad (2)$$

where  $\mathbf{p}' = (x', y')$ ,  $\mathcal{N}(\mathbf{p}')$  denotes the four discrete integer grid locations surrounding the continuous coordinate  $\mathbf{p}'$ , and  $(x_q, y_q)$  represents the coordinates of the discrete grid  $\mathbf{q}$ .

After this warping operation driven by the radar kinematics prior, highly dynamic objects in historical frames are precisely projected onto their corresponding physical locations in the current frame. Subsequently, the aligned historical feature  $\hat{\mathbf{F}}_{\text{cam}}^{t-1}$  and the current-frame camera feature  $\mathbf{F}_{\text{cam}}^t$  are concatenated along the channel dimension and processed by a temporal aggregation module  $\mathcal{F}$ , which consists of a  $1 \times 1$  convolution and Layer Normalization, to produce the temporally enhanced dense representation  $\tilde{\mathbf{F}}_{\text{cam}}^t$ :

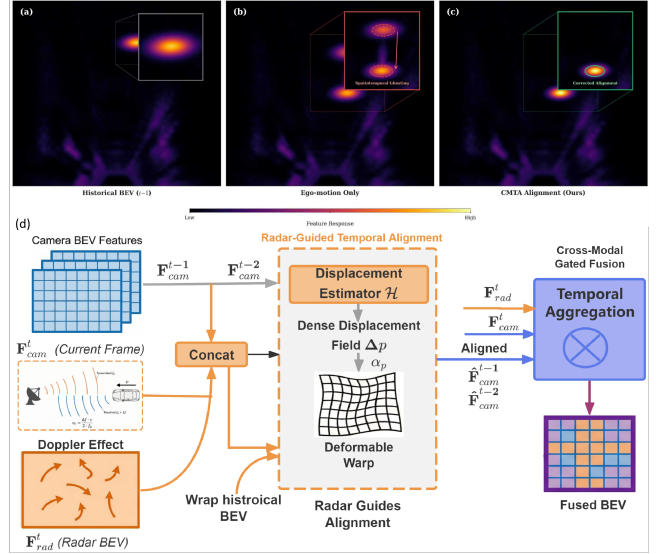
$$\tilde{\mathbf{F}}_{\text{cam}}^t = \mathcal{F}(\text{Concat}(\hat{\mathbf{F}}_{\text{cam}}^{t-1}, \mathbf{F}_{\text{cam}}^t)). \quad (3)$$

To capture long-term dynamic dependencies, the above operations are recursively applied to  $T$  historical frames in a cyclical manner along a time-series sliding window. Ultimately, CMTA constructs a robust multimodal feature representation with strong spatiotemporal consistency, particularly for highly dynamic targets.

### 3.3 Difficulty-Aware Circular Distribution Mechanism (DACD)

Existing query-based 3D detection paradigms typically use uniform Cartesian grids or equidistant polar coordinate radial sampling to initialize object queries. However, such isotropic sampling strategies ignore the highly non-uniform distribution of perception difficulty in autonomous driving scenarios: close-range areas are prone to severe object occlusions, whereas long-range regions suffer from image blur due to decreased optical resolution and sparse radar point clouds. As shown in Figure 4(a), this causes the detection difficulty to surge at both distance extremes, while targets in these regions are under-represented in the training set. To incorporate real-world spatial degradation priors and improve generalizability, we propose a Difficulty-Aware Circular Distribution (DACD) mechanism. Compared to uniform sampling, DACD concentrates more queries in these high-difficulty areas (Figure 4(b)) to improve detection efficiency. First, we conduct a statistical analysis of the baseline model's spatial detection capability on the training set. We partition the radial detection range  $[0, R_{\text{max}}]$  into  $B$  bins, with  $r_k$  denoting the center distance of the  $k$ -th bin. Based on the recall rate  $\text{Recall}(r_k)$  within this interval, we define a discrete empirical detection difficulty prior:

$$D_{\text{emp}}(r_k) = 1 - \text{Recall}(r_k). \quad (4)$$



**Figure 3: Illustration of the proposed Cross-Modal Temporal BEV Alignment (CMTA) module.** (a)-(c) Comparison of temporal alignment results. Ego-motion-only compensation leads to noticeable spatial deviation and ghosting for dynamic objects, whereas CMTA produces better-aligned BEV features. (d) Pipeline of CMTA. Radar Doppler cues and historical camera BEV features are used to estimate a dense displacement field, which warps historical BEV features toward the current frame for temporally consistent aggregation.

To equip the model with cross-scenario generalization capability and decouple it from dataset-specific biases, DACD avoids directly using discrete empirical values, instead parameterizing them into a continuous composite difficulty distribution function  $D(r)$ . Considering the three main physical degradation factors, we formalize  $D(r)$  as:

$$D(r) = w_1 \cdot D_{\text{occ}}(r) + w_2 \cdot D_{\text{blur}}(r) + w_3 \cdot D_{\text{radar}}(r), \quad (5)$$

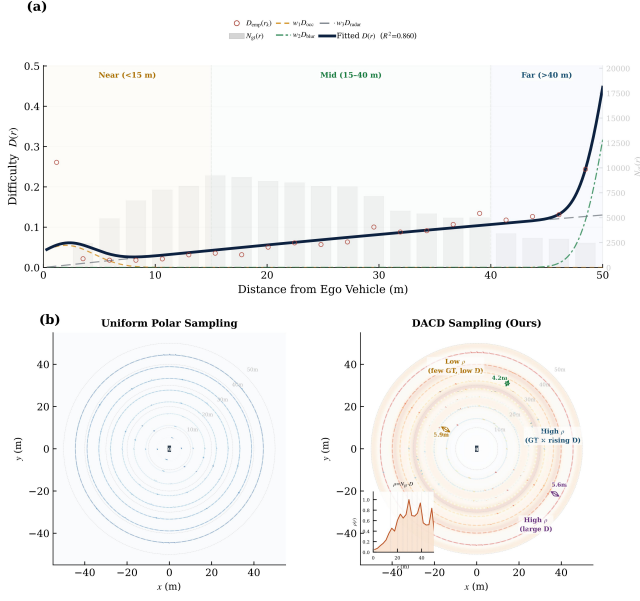
where  $w_1$ ,  $w_2$ , and  $w_3$  are balancing weights. We use Gaussian functions with different means  $\mu$  and variances  $\sigma$  to model visual degradation, and an exponential saturation function to model radar degradation:

$$D_{\text{occ}}(r) = \exp\left(-\frac{(r - \mu_{\text{near}})^2}{2\sigma_{\text{near}}^2}\right), \quad (6)$$

$$D_{\text{blur}}(r) = \exp\left(-\frac{(r - \mu_{\text{far}})^2}{2\sigma_{\text{far}}^2}\right), \quad (7)$$

$$D_{\text{radar}}(r) = 1 - \exp(-\beta r). \quad (8)$$

The parameters  $(\mu_{\text{near}}, \sigma_{\text{near}}, \mu_{\text{far}}, \sigma_{\text{far}}, \beta)$  are obtained via offline curve fitting on  $D_{\text{emp}}(r_k)$ , establishing a continuous difficulty model aligned with physical characteristics. During initialization, DACD organizes  $Q$  query vectors into  $K$  concentric rings, with  $N = Q/K$  queries evenly distributed per ring. The angle  $\theta$  is uniformly distributed, while radial allocation



**Figure 4: Analysis of the Difficulty-Aware Circular Distribution Mechanism (DACD). (a) Distance-aware difficulty modeling for query initialization. The radial importance distribution characterizes the non-uniform detection difficulty across the BEV space. (b) Comparison between uniform sampling and DACD. DACD allocates queries more densely to high-importance regions, leading to more effective use of the query budget.**

is determined by inverse transform sampling. Specifically, incorporating the object distribution prior  $N_{gt}(r)$ , we define the sampling density:

$$\rho(r_k) = N_{gt}(r_k) \cdot D(r_k).$$

We compute its cumulative distribution function (CDF)  $\Phi(r)$  and obtain the ring radii via:

$$\Phi(r) = \frac{\sum_{r_k \leq r} \rho(r_k)}{\sum_{k=1}^B \rho(r_k)}, \quad d_i = \Phi^{-1}\left(\frac{i}{K+1}\right), \quad i = 1, \dots, K. \quad (9)$$

In practice,  $r$  is discretized into  $B$  bins using  $r_k$ , and  $\Phi^{-1}(\cdot)$  is approximated via linear interpolation. Here,  $d_i$  denotes the reference radius of the  $i$ -th query ring. Through this inverse sampling strategy, regions with higher difficulty or denser targets lead to faster CDF growth, resulting in more compact query allocation in these areas.

### 3.4 Dual-Robust Cross-Modal Decoder

After spatial initialization via DACD, Object Queries must anchor targets in 3D space and extract cross-modal features. However, pure vision queries lacking geometric priors, and adverse environments fluctuate modality reliability. To address these vulnerabilities, we propose the Dual-Robust Cross-Modal Decoder. As

shown in Figure 2, the decoder consists of two stages: it sequentially performs semantic-radar calibration and uncertainty-aware gated fusion.

**Phase 1: Semantic and Radar-guided Calibration (SRGC).** For the  $q$ -th query  $\mathbf{f}_q \in \mathbb{R}^D$ , we decode its BEV reference center  $\mathbf{c}_q \in \mathbb{R}^2$  and extract local features from the radar BEV map  $\mathbf{R}_0 \in \mathbb{R}^{C' \times H \times W}$  using bilinear interpolation  $\mathcal{B}(\cdot)$ , followed by projection  $\phi(\cdot)$ :

$$\mathbf{a}_q = \phi(\mathcal{B}(\mathbf{R}_0, \mathbf{c}_q)). \quad (10)$$

The anchor context  $\mathbf{a}_q \in \mathbb{R}^D$  encodes depth and velocity information, providing spatial constraints.

To compensate for the semantic deficiency of radar, we introduce an LLM-based semantic codebook. Specifically, text descriptions of  $K$  object categories are projected into the query space via a frozen text encoder, forming  $\mathbf{E} \in \mathbb{R}^{K \times D}$ . A cross-attention mechanism aggregates category-level priors  $\mathbf{s}_q$ :

$$\mathbf{s}_q = \text{Softmax}\left(\frac{(\mathbf{W}_Q \mathbf{f}_q)(\mathbf{W}_K \mathbf{E})^\top}{\sqrt{D}}\right)(\mathbf{W}_V \mathbf{E}). \quad (11)$$

The calibrated representation is  $\tilde{\mathbf{a}}_q = \mathbf{a}_q + \lambda \cdot \tanh(\text{MLP}(\mathbf{s}_q))$ , which serves as a prior to guide deformable attention sampling. Compared to independent embeddings, the semantic priors encode cross-category physical relationships, providing more stable category and scale information. To maintain training stability, we introduce a gating mechanism to control prior injection:

$$\mathbf{g}_q = \sigma(\mathbf{W}_g[\mathbf{f}_q; \tilde{\mathbf{a}}_q]), \quad \mathbf{f}'_q = \mathbf{f}_q + \mathbf{g}_q \odot \tilde{\mathbf{a}}_q. \quad (12)$$

The enhanced query  $\mathbf{f}'_q$  is then used for deformable attention sampling to obtain image features  $\mathbf{f}_q^{\text{cam}}$  and radar features  $\mathbf{f}_q^{\text{rad}}$ .

**Phase 2: Uncertainty-Aware Cross-Modal Gated Fusion (UACGF).**

After sampling, each query aggregates  $\mathbf{f}_q^{\text{cam}}$  and  $\mathbf{f}_q^{\text{rad}}$ . To address reliability fluctuations under degradation, we introduce modality-specific confidence heads that predict uncertainties  $u_q^{\text{cam}}$  and  $u_q^{\text{rad}}$ , which are normalized into confidence weights:

$$\gamma_q^{\text{cam}} = \frac{\exp(-u_q^{\text{cam}})}{\exp(-u_q^{\text{cam}}) + \exp(-u_q^{\text{rad}})}, \quad \gamma_q^{\text{rad}} = 1 - \gamma_q^{\text{cam}}. \quad (13)$$

The gated feature is computed as  $\mathbf{S}_q = \gamma_q^{\text{cam}} \mathbf{f}_q^{\text{cam}} + \gamma_q^{\text{rad}} \mathbf{f}_q^{\text{rad}}$ , suppressing noise from degraded modalities. The uncertainty branch is trained end-to-end via detection losses, without additional supervision. In practice, this mechanism adaptively adjusts modality weights, preventing single-modality collapse.

We then perform multi-head cross-attention using  $\mathbf{S}_q$  as Query and the original features as Key and Value:

$$\mathbf{h}_q = \text{MHA}(\mathbf{Q} = \mathbf{S}_q, \mathbf{K} = \mathbf{V} = [\mathbf{f}_q^{\text{cam}}; \mathbf{f}_q^{\text{rad}}]). \quad (14)$$

Finally,  $\mathbf{S}_q$  and  $\mathbf{h}_q$  are concatenated and linearly projected to obtain the final query representation:

$$\mathbf{O}_q = \mathbf{W}_o[\mathbf{S}_q; \mathbf{h}_q]. \quad (15)$$

**Table 1: Comparison of 3D object detection results on nuScenes val set. ‘C’ and ‘R’ represent camera and radar, respectively.**

| Method           | Input | Backbone   | Image Size | NDS↑        | mAP↑        | mATE↓ | mASE↓ | mAOE↓ | mAVE↓ | mAAE↓ |
|------------------|-------|------------|------------|-------------|-------------|-------|-------|-------|-------|-------|
| SparseBEV[22]    | C     | ResNet-50  | 256 × 704  | 54.5        | 43.2        | 0.606 | 0.274 | 0.387 | 0.251 | 0.186 |
| StreamPETR[38]   | C     | ResNet-50  | 256 × 704  | 55.0        | 45.0        | 0.613 | 0.267 | 0.413 | 0.265 | 0.196 |
| CRN[13]          | C+R   | ResNet-50  | 256 × 704  | 56.0        | 49.0        | 0.487 | 0.277 | 0.542 | 0.344 | 0.197 |
| RCBEVDet++[21]   | C+R   | ResNet-50  | 256 × 704  | 60.4        | 51.9        | 0.488 | 0.268 | 0.408 | 0.221 | 0.177 |
| RacFormer[6]     | C+R   | ResNet-50  | 256 × 704  | 61.3        | 54.1        | 0.478 | 0.261 | 0.449 | 0.208 | 0.180 |
| RCTDistill[2]    | C+R   | ResNet-50  | 256 × 704  | 62.2        | 55.2        | 0.434 | 0.241 | 0.432 | 0.263 | 0.174 |
| SPRFusion (Ours) | C+R   | ResNet-50  | 256 × 704  | <b>62.4</b> | <b>55.4</b> | 0.482 | 0.263 | 0.376 | 0.214 | 0.189 |
| SparseBEV[22]    | C     | ResNet-101 | 512 × 1408 | 59.2        | 50.1        | 0.562 | 0.265 | 0.321 | 0.243 | 0.195 |
| StreamPETR[38]   | C     | ResNet-101 | 512 × 1408 | 59.2        | 50.4        | 0.569 | 0.262 | 0.315 | 0.257 | 0.199 |
| CRN[13]          | C+R   | ResNet-101 | 512 × 1408 | 59.2        | 52.5        | 0.460 | 0.273 | 0.443 | 0.352 | 0.180 |
| HyDRa[40]        | C+R   | ResNet-101 | 512 × 1408 | 61.7        | 53.6        | 0.416 | 0.264 | 0.407 | 0.231 | 0.186 |
| RacFormer[6]     | C+R   | ResNet-101 | 512 × 1408 | 63.0        | 57.3        | 0.476 | 0.261 | 0.428 | 0.213 | 0.183 |
| RCTDistill[2]    | C+R   | ResNet-101 | 512 × 1408 | 65.5        | 59.0        | 0.378 | 0.250 | 0.346 | 0.236 | 0.186 |
| SPRFusion (Ours) | C+R   | ResNet-101 | 512 × 1408 | <b>65.9</b> | <b>59.3</b> | 0.402 | 0.268 | 0.308 | 0.204 | 0.197 |

**Table 2: Comparison on the nuScenes test set. The VoVNet-99 (V2-99) is pre-trained from DD3D with extra data. ‘L’, ‘C’, and ‘R’ represent LiDAR, camera, and radar, respectively. We do not use future frames, test-time data augmentation, or model ensembling**

| Method           | Input | Backbone | Image Size | NDS↑        | mAP↑        | mATE↓ | mASE↓ | mAOE↓ | mAVE↓ | mAAE↓ |
|------------------|-------|----------|------------|-------------|-------------|-------|-------|-------|-------|-------|
| CenterPoint[49]  | L     | VoxelNet | -          | 67.3        | 60.3        | 0.262 | 0.239 | 0.361 | 0.288 | 0.136 |
| MVFusion[42]     | C+R   | V2-99    | 640 × 1600 | 51.7        | 45.3        | 0.569 | 0.246 | 0.379 | 0.781 | 0.128 |
| BEVDepth[17]     | C     | ConvX-B  | 640 × 1600 | 60.9        | 52.0        | 0.445 | 0.243 | 0.352 | 0.347 | 0.128 |
| CRN[13]          | C+R   | ConvX-B  | 640 × 1600 | 62.4        | 57.5        | 0.416 | 0.264 | 0.456 | 0.365 | 0.130 |
| RayFormer[7]     | C     | V2-99    | 640 × 1600 | 63.3        | 55.5        | 0.507 | 0.245 | 0.326 | 0.247 | 0.123 |
| SparseBEV[22]    | C     | V2-99    | 640 × 1600 | 63.6        | 55.6        | 0.485 | 0.244 | 0.332 | 0.246 | 0.117 |
| RCBEVDet[21]     | C+R   | V2-99    | 640 × 1600 | 63.9        | 55.0        | 0.390 | 0.234 | 0.362 | 0.259 | 0.113 |
| HyDRa[40]        | C+R   | V2-99    | 640 × 1600 | 64.2        | 57.4        | 0.398 | 0.251 | 0.423 | 0.249 | 0.122 |
| RCTrans[18]      | C+R   | V2-99    | 640 × 1600 | 64.7        | 57.8        | 0.459 | 0.245 | 0.392 | 0.198 | 0.121 |
| CRTFusion[35]    | C+R   | ConvX-B  | 512 × 1408 | 64.9        | 58.3        | 0.365 | 0.261 | 0.405 | 0.262 | 0.132 |
| RacFormer[6]     | C+R   | V2-99    | 640 × 1600 | 65.9        | 59.2        | 0.407 | 0.244 | 0.345 | 0.238 | 0.132 |
| SpaRc[41]        | C+R   | V2-99    | 640 × 1600 | 67.1        | 60.0        | -     | -     | -     | -     | -     |
| RCTDistill[2]    | C+R   | ConvX-B  | 512 × 1408 | 67.4        | 60.3        | 0.334 | 0.245 | 0.334 | 0.251 | 0.113 |
| SPRFusion (Ours) | C+R   | V2-99    | 640 × 1600 | <b>68.2</b> | <b>62.9</b> | 0.314 | 0.246 | 0.364 | 0.266 | 0.137 |

## 4 Experiments

### 4.1 Datasets and Metrics

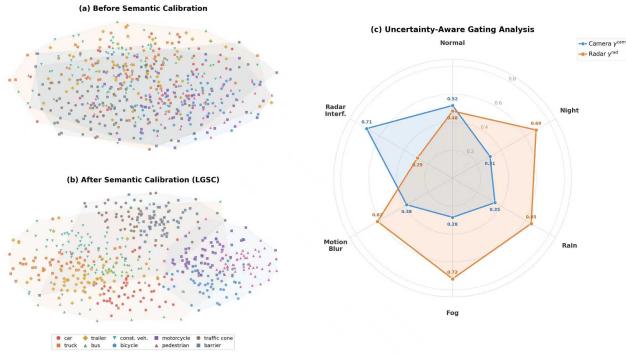
We evaluate SPRFusion on two public autonomous driving benchmarks: nuScenes [4] and View-of-Delft (VoD) [31].

**nuScenes.** The dataset contains 1,000 driving scenes, split into 700/150/150 scenes for training, validation, and testing, respectively. Each scene lasts around 20 seconds, and key samples are annotated at 2 Hz. The nuScenes evaluation metrics include Average Translation Error (ATE), Average Scale Error (ASE), Average Orientation Error (AOE), Average Velocity Error (AVE), and Average Attribute Error (AAE). Our evaluation follows the official nuScenes benchmark metrics and primarily validates the performance of our model on mean accuracy (mAP) and nuScenes Detection Score (NDS).

**VoD.** The dataset contains 8,693 synchronized radar-camera-LiDAR frames captured in urban driving environments. The dataset provides 123,106 3D bounding box annotations for cars, pedestrians, and cyclists. Following prior work, we report 3D detection mAP on both the Region of Interest (RoI) and the Entire Annotated Area (EAA).

### 4.2 Implementation Details

For the nuScenes dataset, we initialize 900 queries, while for the VoD dataset, we initialize 600 queries. Our model is trained using the AdamW optimizer [26] with a global batch size of 16. The backbone starts the learning process with a learning rate of 4e-5, while applying a cosine annealing [25] strategy for rate adjustment. For the image feature encoding backbone, we employ the



**Figure 5: Analysis of the two robustness mechanisms in the proposed decoder.**(a) Feature distribution before semantic calibration. (b) Feature distribution after semantic calibration, where category features become more compact and better separated. (c) Uncertainty-aware gating under different degradation conditions, showing adaptive reweighting between camera and radar according to modality reliability.

standard networks ResNet-50 [10], ResNet-101 [10], and VoVNet-99 (V2-99) [16]. The ResNet parameters are pre-trained on nuIm-ages [12], while the V2-99 parameters are pre-trained on DD3D [32]. All training is conducted on 4 NVIDIA A800 GPUs.

### 4.3 Main Results

**NuScenes results.** We first compare SPRFusion with state-of-the-art 3D detectors on the nuScenes benchmark. Tables 1 and 2 report the results on the validation and test sets, respectively. On the validation set, under the ResNet-50 setting with an input resolution of  $256 \times 704$ , SPRFusion achieves 62.4 NDS and 55.4 mAP, outperforming strong radar-camera baselines such as RCBEVDet++, RaCFormer, and RCTDistill. With a stronger ResNet-101 backbone and a higher input resolution of  $512 \times 1408$ , SPRFusion further improves to 65.9 NDS and 59.3 mAP, again establishing the best overall performance among the compared radar-camera methods. On the official test set, SPRFusion achieves 68.2 NDS and 62.9 mAP with a VoVNet-99 backbone at  $640 \times 1600$ , surpassing previous radar-camera fusion approaches including RaCFormer, RCTrans, and RCTDistill. These results demonstrate that SPRFusion generalizes effectively across different backbone configurations and remains highly competitive even against strong camera-only and LiDAR-based detectors.

**VoD results.** We further evaluate SPRFusion on the View-of-Delft benchmark, and the results are summarized in Table 3. SPRFusion achieves 79.87 mAP on the RoI benchmark and 59.32 mAP on the EAA benchmark, outperforming previous radar-camera fusion methods such as RCBEVDet, RaCFormer, and HGSFusion. The consistent gains on VoD indicate that the proposed framework generalizes well beyond nuScenes and remains effective under different sensor layouts and evaluation protocols.

**Table 3: Comparison of 3D object detection results on VoD val set. RoI: Region of Interest, EAA: Entire Annotated Area.**

| Methods            | Input | ROI          | EAA          |
|--------------------|-------|--------------|--------------|
| PointPillars[14]   | R     | 67.48        | 45.18        |
| RadarPillarNet[31] | R     | 65.86        | 46.01        |
| BEVFusion[23]      | C+R   | 68.52        | 49.25        |
| RCFusion[19]       | C+R   | 69.23        | 49.65        |
| RCBEVDet[21]       | C+R   | 69.80        | 49.99        |
| RacFormer[6]       | C+R   | 78.57        | 54.44        |
| LXL[45]            | C+R   | 72.93        | 56.31        |
| HGSFusion[9]       | C+R   | 79.46        | 58.96        |
| SPRFusion (Ours)   | C+R   | <b>79.87</b> | <b>59.32</b> |

**Table 4: Performance comparison under different weather and lighting conditions.**

| Methods         | Input | Rainy       | Day         | Night       |
|-----------------|-------|-------------|-------------|-------------|
| CenterPoint[49] | L     | 59.2        | 62.8        | 35.4        |
| RayFormer[7]    | C     | 49.1        | 45.7        | 27.3        |
| SPRFusion       | C+R   | <b>56.8</b> | <b>57.4</b> | <b>34.3</b> |

**Table 5: Ablation of the main components of SPRFusion.**Each component improves the 3D detection performance consistently.

| Model Configuration | Input | NDS↑                    | mAP↑                    |
|---------------------|-------|-------------------------|-------------------------|
| Baseline            | C     | 50.4                    | 40.3                    |
| + CMTA              | C+R   | 53.6 <b>↑3.1</b>        | 44.5 <b>↑4.2</b>        |
| + DACD              | C+R   | 54.4 <b>↑0.8</b>        | 45.4 <b>↑0.9</b>        |
| + SRGC              | C+R   | 56.1 <b>↑1.7</b>        | 47.4 <b>↑2.0</b>        |
| + UACGF             | C+R   | <b>57.3</b> <b>↑1.2</b> | <b>48.2</b> <b>↑0.8</b> |

**Performance under challenging conditions.** To further verify the robustness of SPRFusion, we evaluate its performance under adverse weather and illumination conditions. As shown in Table 4, SPRFusion consistently outperforms representative camera-only and LiDAR-based baselines in rainy and nighttime scenarios. In particular, the gains under rain and low-light conditions support our central claim that the proposed semantic- and physics-driven fusion strategy improves resistance to modal degradation and environmental uncertainty.

### 4.4 Ablation Studies

Unless otherwise specified, all ablation experiments are conducted on the nuScenes validation set using a ResNet-50 image backbone and an input resolution of  $256 \times 704$ .

**Major components.** We first study the contribution of each major component of SPRFusion. Starting from a query-based radar-camera baseline with ego-motion-based rigid alignment, uniform query initialization, and simple feature concatenation, we progressively add CMTA, DACD, semantic and radar-guided calibration, and UACGF. As shown in Table 5, the full model improves the baseline from 50.4 to 57.3 NDS and from 40.3 to 48.2 mAP,

demonstrating that all proposed components contribute positively and work synergistically.

**Table 6: Analysis of detecting moving objects .It evaluates the effectiveness of the implicit dynamic catcher on objects with different motion states.**

| Temporal Alignment Strategy | 0 m/s       | (0, 5] m/s  |              | >5 m/s      |              |
|-----------------------------|-------------|-------------|--------------|-------------|--------------|
|                             | mAP         | mAP         | mAVE         | mAP         | mAVE         |
| Temporal-free               | 46.1        | 13.4        | 0.742        | 30.1        | 0.806        |
| Ego-motion                  | 50.1        | 19.6        | 0.543        | 42.1        | 0.712        |
| CMTA                        | <b>53.6</b> | <b>22.6</b> | <b>0.497</b> | <b>47.1</b> | <b>0.652</b> |

**Effect of CMTA on Dynamic Object Detection** We next evaluate the effectiveness of CMTA on dynamic object detection. Table 6 compares different temporal alignment strategies over objects with different motion states. Compared with temporal-free and ego-motion-based alignment, CMTA yields the largest gains on moving objects, especially in the fast-moving group, where mAP increases from 42.1 to 47.1 and mAVE decreases from 0.712 to 0.652. These results verify the effectiveness of Doppler-guided non-rigid temporal warping for dynamic alignment.

**Table 7: Ablation of Semantic and Radar-guided Calibration. We employ Average Precision (AP) to evaluate the performance of individual categories.**

| Radar-guided Sampling | Semantic Calibration | mAP         | NDS         | mATE↓ | mASE↓ | Ped  | Bicycle | FPS  |
|-----------------------|----------------------|-------------|-------------|-------|-------|------|---------|------|
| ✗                     | ✗                    | 45.8        | 53.9        | 0.576 | 0.252 | 53.2 | 47.2    | 20.6 |
| ✓                     | ✗                    | 46.2        | 54.3        | 0.506 | 0.250 | 53.5 | 47.3    | 20.5 |
| ✗                     | ✓                    | 46.4        | 54.6        | 0.564 | 0.226 | 54.4 | 48.0    | 20.1 |
| ✓                     | ✓                    | <b>47.6</b> | <b>55.4</b> | 0.496 | 0.214 | 54.6 | 48.4    | 19.8 |

**Table 8: Ablation of initialization strategies. AP values are evaluated across different distance ranges: extreme near-field (< 15m), mid-field (15 – 40m), and far-field (> 40m).**

| InitializationStrategy            | mAP         | NDS         | AP    |             |             |
|-----------------------------------|-------------|-------------|-------|-------------|-------------|
|                                   |             |             | < 15m | 15 – 40m    | > 40m       |
| (a) Cartesian Uniform             | 43.1        | 51.6        | 44.6  | 43.5        | 34.4        |
| (b) Polar Isometric (RayFormer)   | 43.8        | 52.4        | 45.2  | 44.2        | 33.2        |
| (c) Linear Increasing (RaCFormer) | 44.2        | 52.8        | 43.6  | 44.5        | 35.8        |
| (d) DACD (Ours)                   | <b>44.8</b> | <b>53.5</b> | 42.8  | <b>45.6</b> | <b>38.0</b> |

**Effect of semantic and radar-guided calibration.** Table 7 analyzes the two components in the calibration stage. Radar-guided sampling mainly improves localization quality, reducing mATE from 0.576 to 0.506. Semantic calibration mainly enhances semantic discrimination and scale estimation, reducing mASE to 0.226 while also improving pedestrian and bicycle AP. Combining both modules yields the best overall performance, achieving 47.6 mAP and 55.4 NDS with only minor inference overhead.

**Table 9: Ablation of fusion strategies with uncertainty estimation and gating mechanism. We validate the robustness of the proposed modules by evaluating their performance under adverse weather (rainy) and low-illumination (night-time) conditions.**

| Fusion Strategy | Uncertainty Estimation | Gating Mechanism | mAP         | NDS         | mAP <sub>rain</sub> | mAP <sub>night</sub> |
|-----------------|------------------------|------------------|-------------|-------------|---------------------|----------------------|
| Concat          | ✗                      | ✗                | 45.2        | 52.1        | 44.6                | 34.2                 |
| Concat          | ✓                      | ✗                | 45.4        | 52.6        | 44.8                | 35.1                 |
| Concat          | ✗                      | ✓                | 45.3        | 52.4        | 44.6                | 34.6                 |
| UACGF           | ✓                      | ✓                | <b>46.5</b> | <b>53.8</b> | <b>45.6</b>         | <b>35.8</b>          |

**Effect of DACD.** We further compare DACD with several representative query initialization strategies. As shown in Table 8, DACD achieves the best overall mAP and NDS under the same query budget, while also delivering the strongest performance in the mid-range and far-range regions. In contrast, linearly increasing radial density mainly favors far-range detection at the expense of near-to-mid range accuracy. These results validate the effectiveness of difficulty-aware non-uniform query allocation.

**Effect of UACGF.** Finally, Table 9 evaluates the proposed uncertainty-aware gated fusion module. Compared with direct concatenation, adding uncertainty estimation or gating alone yields only modest gains, whereas their combination achieves the best performance. More importantly, the improvement becomes more pronounced under rainy and nighttime conditions, confirming that UACGF can dynamically adjust modal reliability and improve robustness in degraded environments.

## 5 Conclusion

In this paper, we present SPRFusion, a novel Radar-Camera fusion framework designed to fully exploit the synergistic information from both radar and camera sensors for efficient 3D object detection. Specifically, we propose a query-based multi-modal fusion approach that fully leverages the synergistic advantages and complementarity of radar and camera sensors. Our framework incorporates difficulty-aware polar query initialization, semantic- and radar-guided reliable sampling, Uncertainty-Aware Cross-Modal Gated Fusion (UACGF), and radar-based cross-modal temporal alignment. SPRFusion achieves state-of-the-art (SOTA) performance on the nuScenes validation and test sets, as well as the VoD dataset. Notably, it demonstrates exceptional robustness in challenging scenarios such as adverse weather (rainy) and low-illumination (nighttime) conditions. In conclusion, SPRFusion represents a significant advancement in multi-modal perception for autonomous driving. We envision that this work will provide a novel paradigm and serve as a compelling reference for future research in 3D object detection.

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